Vivek Raj

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Professional Summary

Final-year Electronics and Telecommunication Engineering student and software developer with hands-on experience building full-stack, backend and data-driven applications. Experienced with React, Node.js, Express, FastAPI, PostgreSQL, Redis and MySQL, with a strong foundation in REST API design, authentication, database systems and scalable application architecture. Interested in software engineering, AI/ML and developing reliable products that solve practical problems.

Skills

Programming Languages: C++, C, Python, JavaScript, TypeScript, SQL
Frontend Development: React.js, HTML5, CSS3, Tailwind CSS, Vite, React Router
Backend & APIs: Node.js, Express.js, FastAPI, REST APIs, JWT Authentication, Role-Based Access Control
Databases & Caching: PostgreSQL, MySQL, MongoDB, SQLite, Redis
Cloud & DevOps: AWS, Docker, Kubernetes, Vercel, Render
Developer Tools: Git, GitHub, VS Code, Postman, MySQL Workbench, Linux
Data & AI/ML: Pandas, NumPy, Scikit-learn, Machine Learning, RAG, LLM APIs
Relevant Coursework: Data Structures & Algorithms, DBMS, OOP, Operating Systems, Computer Networks

Projects

DevPulse - Full-Stack Developer Platform | React · Node.js · Express · PostgreSQL · Redis

- Engineered a full-stack project-management platform using a layered routes, controllers, services and repositories architecture to keep backend logic modular and maintainable.
- Designed REST APIs for project creation, retrieval and management with JWT authentication, role-based access control, request validation, pagination and structured error handling.
- Integrated PostgreSQL persistence with Redis caching using TTL, namespace validation, single-flight protection and database fallback; developed 174 automated tests for API and service behavior.

PAIMANA - Infrastructure Monitoring Platform | TypeScript · FastAPI · PostgreSQL *Smart India Hackathon 2026*

- Built an infrastructure-monitoring platform with project exploration, dashboards, risk views and role-based access for four user roles.
- Designed PostgreSQL data models for ingestion batches, monthly snapshots, milestones and audit events around a 177,679-row source dataset.
- Developed data-ingestion and transformation workflows and prepared project-cost and schedule-prediction model integrations for risk analysis.

Gambling Management System - DBMS Full-Stack Application | React · Node.js · Express · MySQL

- Developed a role-based application for Admin, Dealer, Cashier and Player workflows with authentication and persistent database-backed operations.
- Implemented user, game, table, wallet and betting workflows while demonstrating primary/foreign keys, CHECK constraints, JOINs, views, triggers and SQL transactions.
- Connected frontend actions to backend APIs and relational persistence to demonstrate complete end-to-end DBMS functionality.

Education

Sardar Patel Institute of Technology (S.P.I.T.), Mumbai *2024 - 2028*
B.Tech in Electronics and Telecommunication Engineering - Pursuing (Final Year)
HSC (12th): 87% - Matoshri Junior College, Nashik | ICSE (10th): 97.89% - Ashoka Universal School, Nashik

Leadership & Campus Involvement

- Head of Creative Department, Rotaract Club of S.P.I.T. - led creative planning and execution for club campaigns and events, coordinated team deliverables, delegated design/content responsibilities and maintained consistent visual communication across initiatives.
- Head of Creative, IETE S.P.I.T. - managed creative direction for technical events and committee activities, collaborating with event and outreach teams on promotional campaigns, announcements and event communication.

Competitive Programming

Codeforces: 1050 Rating | CodeChef: 2-Star